

eSim Summer Fellowship 2026

Task 4: Installation Bug Report on Ubuntu 25.04

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Repository	https://github.com/yousseufkeita-git/eSim

1. Introduction

This report documents the bugs encountered and fixes applied during the installation of eSim 2.5 on Ubuntu 25.04 LTS (Plucky Puffin), as part of the FOSSEE Summer Fellowship 2026 Task 4. The objective was to identify compatibility issues, analyze root causes, fix at least one issue, and produce a detailed technical report.

eSim is an open-source EDA tool developed by the FOSSEE team at IIT Bombay, built on KiCad, Ngspice, and GHDL. As Ubuntu 25.04 is a new release, several dependency and compatibility issues were expected — this report documents all findings systematically.

2. Environment Setup

2.1 System Configuration

Host OS	Windows 11 (HP EliteBook)
RAM	32 GB
Virtualization	WSL2 (Windows Subsystem for Linux)
Guest OS	Ubuntu 25.04 (Plucky Puffin)
eSim Version	eSim 2.5 (branch: installers)
Python Version	Python 3.13.3

2.2 Ubuntu 25.04 Installation on WSL2

Ubuntu 25.04 is not natively available in the WSL Store (confirmed via `wsl --list --online`). The installation was performed manually by downloading the official Ubuntu 25.04 cloud rootfs image and importing it into WSL2:

```
# Download Ubuntu 25.04 rootfs
curl.exe --output ubuntu2504.tar.xz https://cloud-images.ubuntu.com/releases/plucky/release/ubuntu-25.04-server-cloudimg-amd64-root.tar.xz

# Import into WSL2
wsl --import Ubuntu2504 C:\WSL\Ubuntu2504 ubuntu2504.tar.xz

# Launch the distro
wsl -d Ubuntu2504
```

Note: No dedicated WSL rootfs (.wsl file) exists for Ubuntu 25.04 on the Ubuntu cloud images server, unlike previous LTS versions. This is an additional environment-level issue worth noting.

2.3 Repository Setup

The eSim repository was cloned from GitHub and checked out to the installers branch:

```
git clone https://github.com/FOSSEE/eSim.git
cd eSim
git checkout installers
cd Ubuntu
./install-eSim.sh --install
```

Note: The task documentation references a branch named 'installer', but the actual branch name in the repository is 'installers' (with an 's'). This discrepancy may confuse new contributors.

3. Bug Summary

A total of 4 bugs were identified during the installation process. 3 bugs were successfully fixed; 1 bug was identified and documented with a proposed solution.

Bug #	Description	File	Severity	Status
#1	Ubuntu 25.04 not recognized	install-eSim.sh	Critical	Fixed
#2	Missing 'install' in apt-get command	install-eSim-24.04.sh	Medium	Fixed
#3	KiCad 6.0 PPA incompatible with Ubuntu 25.04	install-eSim-24.04.sh	Critical	Fixed

#4	libgit2-1.8 unavailable on Ubuntu 25.04	install-eSim-24.04.sh	Critical	Identified
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4. Detailed Bug Reports

Bug #1 — Ubuntu 25.04 Not Recognized by Installer	
Severity	Critical
Status	Fixed
File	Ubuntu/install-eSim.sh
Line	Line 30 — case statement in run_version_script()
Error Message	Unsupported Ubuntu version: 25.04 ()
Root Cause	The case statement in install-eSim.sh only handles Ubuntu versions 22.04, 23.04, and 24.04. Ubuntu 25.04 falls into the default (*) branch which prints an error and exits immediately.
Fix Applied	Added a '25.04' case to the switch statement, pointing to install-eSim-24.04.sh as the closest compatible script.
Impact	Installation aborts immediately without attempting any package installation. This is the first critical blocker encountered.

Before fix:

```
        "24.04")
            SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
            ;;
        *)
            echo "Unsupported Ubuntu version: $VERSION_ID
($FULL_VERSION) "
            exit 1
            ;;
```

After fix:

```
        "24.04")
            SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
            ;;
        "25.04")
            SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
            ;;
        *)
```

```

        echo "Unsupported Ubuntu version: $VERSION_ID
($FULL_VERSION) "
        exit 1
    ;;

```

Bug #2 — Missing 'install' Subcommand in apt-get Call	
Severity	Medium
Status	Fixed
File	Ubuntu/install-eSim-scripts/install-eSim-24.04.sh
Line	Line 252
Error Message	E: Invalid operation xz-utils
Root Cause	The script calls 'sudo apt-get xz-utils' without the 'install' subcommand and without the -y flag. apt-get interprets 'xz-utils' as an invalid operation, causing the installation to abort.
Fix Applied	Changed 'sudo apt-get xz-utils' to 'sudo apt-get install -y xz-utils'.
Impact	Installation fails when trying to install the xz-utils compression tool, which is required for handling .tar.xz archives used later in the installation process.

Before fix:

```
sudo apt-get xz-utils
```

After fix:

```
sudo apt-get install -y xz-utils
```

Bug #3 — KiCad 6.0 PPA Does Not Support Ubuntu 25.04	
Severity	Critical
Status	Fixed
File	Ubuntu/install-eSim-scripts/install-eSim-24.04.sh
Line	Line 166
Error Message	404 Not Found — https://ppa.launchpadcontent.net/kicad/kicad-6.0-releases/ubuntu/plucky/Release
Root Cause	The script adds the KiCad 6.0 PPA (kicad/kicad-6.0-releases) to the apt sources. However, this PPA does not publish packages for Ubuntu 25.04 (Plucky), resulting in a 404 error. Ubuntu 25.04 support was

	never added to the KiCad 6.0 PPA.
Fix Applied	Changed kicadppa value from 'kicad/kicad-6.0-releases' to 'kicad/kicad-8.0-releases', which officially supports Ubuntu 25.04. Also removed the stale PPA entry using: <code>sudo add-apt-repository --remove ppa:kicad/kicad-6.0-releases -y</code>
Impact	KiCad cannot be installed at all, blocking the core functionality of eSim. This is the most impactful bug as KiCad is the schematic editor at the heart of eSim.

Before fix:

```
kicadppa="kicad/kicad-6.0-releases"
```

After fix:

```
kicadppa="kicad/kicad-8.0-releases"
```

Additionally, the stale PPA registered from a previous failed attempt was removed manually:

```
sudo add-apt-repository --remove ppa:kicad/kicad-6.0-releases -y
```

Bug #4 — KiCad 8.0 Depends on libgit2-1.8 — Unavailable on Ubuntu 25.04	
Severity	Critical
Status	Identified
File	Ubuntu/install-eSim-scripts/install-eSim-24.04.sh
Line	KiCad 8.0 package dependency
Error Message	kicad : Depends: libgit2-1.8 (>= 1.8.0) but it is not installable
Root Cause	KiCad 8.0 (version 8.0.9-0~ubuntu25.04.1) depends on libgit2-1.8, but Ubuntu 25.04 ships libgit2-1.9. The KiCad PPA package was built against libgit2-1.8 which was available in Ubuntu 24.10, but Ubuntu 25.04 jumped directly to libgit2-1.9. The two versions are not co-installable.
Fix Applied	This bug requires upstream action from the KiCad PPA maintainers to rebuild KiCad 8.0 packages against libgit2-1.9. A workaround would be to compile libgit2-1.8 from source or to use the KiCad snap package instead.
Impact	Even with Bug #3 fixed, KiCad cannot be installed due to this unresolvable dependency conflict. This is an upstream packaging issue that requires coordination between the KiCad PPA team and Canonical.

Available libgit2 versions on Ubuntu 25.04:

```
$ apt-cache search libgit2
libgit2-1.9 - low-level Git library
libgit2-dev - low-level Git library (development files)
```

Proposed solution (requires upstream fix):

- Option 1: Wait for KiCad PPA to publish a rebuild against libgit2-1.9
 - Option 2: Install KiCad via snap: `sudo snap install kicad`
 - Option 3: Compile libgit2-1.8.0 from source and install manually
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5. Recommendations

Based on the bugs identified, the following improvements are recommended for the eSim installer:

1. Add Ubuntu 25.04 (Plucky) support explicitly in `install-eSim.sh` and create a dedicated `install-eSim-25.04.sh` script.
 2. Fix the missing 'install' subcommand on line 252 of `install-eSim-24.04.sh`.
 3. Coordinate with the KiCad PPA team to rebuild KiCad packages against libgit2-1.9 for Ubuntu 25.04 compatibility.
 4. Add a PPA cleanup step in the installer script to remove stale PPA entries before adding new ones.
 5. Update documentation to clarify that the correct branch name is 'installers' (not 'installer') and provide Ubuntu 25.04 WSL setup instructions.
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6. Conclusion

This investigation successfully identified 4 bugs in the eSim 2.5 installer on Ubuntu 25.04, of which 3 were fixed and 1 was thoroughly documented with proposed solutions. The most critical finding is the libgit2-1.8 dependency conflict (Bug #4), which requires upstream intervention from the KiCad PPA maintainers.

The fixes applied in this report bring the installation significantly closer to completion on Ubuntu 25.04 and provide a clear roadmap for full compatibility. All modified files have been committed and pushed to the forked repository at <https://github.com/yousseufkeita-git/eSim> on the 'installers' branch.